BorgWarner

App Note –Line Contactor and Pre-charge Configuration

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1 Introduction

1.1 Purpose and Scope

This document explains detailed configuration for line contactor and pre-charge devices for Genus products, including Dragon8 and HVLP.

1.2 Related documents

Ref	Doc Ref	Title	Revision
1		AppNote – BorgWarner I-protocol	
2		AppNote – Mapping J1939 Items to VehApp	

Table 1. Related Documents

1.3 Document Conventions and Terminology

Typeface	Indicates
WARNING	Very important information for the document user. Use this to indicate instructions or constraints the document user must comply with to prevent serious problems.
CAUTION	Important information for the document user. Use this to indicate instructions or constraints the document user must comply with to prevent undesirable problems.
NOTE	Useful information for the document user.

Table 2. Typographical Conventions

1.4 Glossary

Term	Description

Table 3. Glossary

1.5 Conditions for document usage

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2 Overview of the System

As with Gen4, Genus products can be configured to use a line contactor, and are able to control a pre-charge circuit. However, the configuration and operation of these devices has changed slightly from Gen4. Hopefully the changes are an improvement and simplification over the Gen4 scheme. However, some consideration does need to be paid to their configuration as, although simplified, there is now scope for more flexibility in the configuration.

Some of the major changes from Gen4:

- Some variants of Dragon8 contains an integrated line contactor, more commonly referred to as a battery switch. This device can be controlled using I-Protocol, using a “virtual” analogue output. While Dragon8 contains a number of real physical outputs, there is an additional one for controlling the battery switch.
- Some variants of Dragon8 also includes a pre-charge circuit. Gen4 also included this, but in Dragon8 it is controlled via another virtual output. This means it needs to be mapped to BattApp, but it does offer more flexibility in terms of allowing this function to be controlled by external VCUs.
- It is possible to control multiple line contactors and pre-charge circuits. This is required if using multiple Dragon8 controllers on a vehicle, as they all need to be instructed to control their battery switch.
- BattApp has been modified to check the capacitor voltage from all nodes in order to detect line contactor and battery switch problems, and not just its local capacitor voltage. It also has more intelligent management of systems that don’t protect all controllers on a vehicle with a line contactor. For example, pump and steering controllers might not be isolated.

The following sections in this app note describe the principles behind the operation of the system, and how they are intended to be used together. The system is very flexible, so it is advised that these principles are understood, as this will lead to being able to set up any combination of line contactor and pre-charge protected systems.

The following sections will explain:

- How to use I-Protocol to control the battery switch and pre-charge circuits on a controller, particularly Dragon8
- How to map control of the line contactor and pre-charge circuits to BattApp
- How BattApp controls these devices, and how BattApp expects the system to behave

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3 Controlling the Integrated Battery Switch and Pre-charge Devices (only some variants of Dragon8)

As already mentioned, some variants of Dragon8 features an integrated battery switch and pre-charge circuit. The hardware for these devices are contained “in the box”, but they can be controlled using I-Protocol in the same way as any other device that is connected to the controller.

I-Protocol is a proprietary Sevcon protocol used for control and monitoring of IO modules. It is designed to be compatible with J1939 networks, and allocates a number of PGNs for the control of analogue outputs, and monitoring of analogue and digital inputs. The current revision of the protocol provides the ability to control up to 8 analogue outputs. Dragon8 only has two physical contactor drives, which are controlled by signals OP1 and OP2 in the protocol. However, OP7 and OP8, while not attached to a physical output, represent virtual outputs that control the pre-charge circuit and battery switch respectively.

Signals OP3, OP4, OP5 and OP6 are ignored on Dragon8, but might be used on other products.

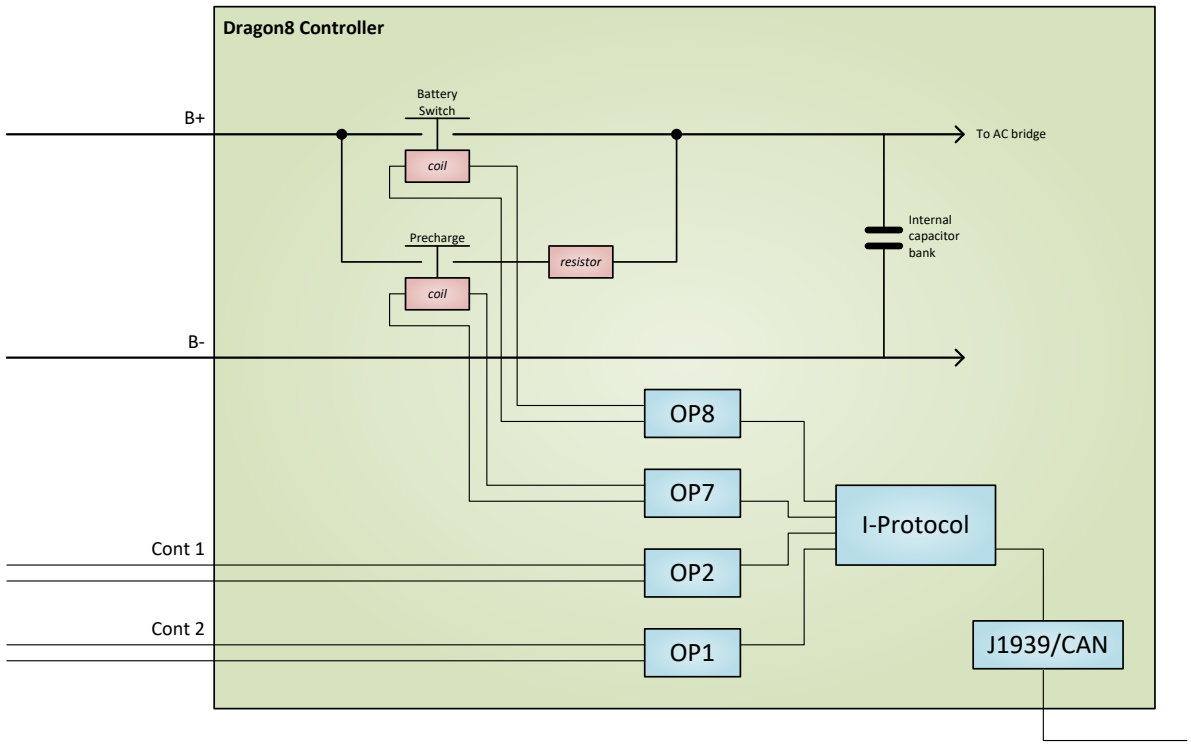


Figure 1 - Block diagram showing how I-Protocol can be used to drive internal battery switch and pre-charge circuits

Figure 1 shows a simplified overview of the connection between the outputs from I-Protocol and the devices they control. The advantage to this approach is that for BattApp, it is simply controlling another output via I-Protocol that is believes is connected to a line contactor. That is to say, the behaviour of BattApp doesn’t change, regardless of whether it is controlling the internal battery switch, or a real external physical contactor.

For the pre-charge circuit, this is also controlled via I-Protocol, and can be turned on and off via CAN. In Gen4, this was controlled using an SDO message. However, for Genus, the SDO message is no longer used, and a specific output has been created for the pre-charge signal, which is controlled by BattApp. Both the line contactor and pre-charge signals from BattApp should be mapped as appropriate to the Dragon8 virtual outputs.

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3.1 Control of Virtual Outputs

As noted already, OP7 and OP8 are virtual outputs for control of the pre-charge and battery switch. In terms of I-Protocol, these are effectively analogue outputs, and are controlled using a 12-bit value. However, in reality, their function is digital, so the actual value that is transmitted is interpreted as described below.

For signal OP8:

- If a value of less than or equal to 0x005 is received, this will open the battery switch, subject to the conditions outlined in section 3.2.
- If a value greater than 0x005 and less than 0xFFF is received, this will close the battery switch, subject to the conditions outlined in section 3.2.
- If the value 0xFFF is received, the state of the battery switch remains unchanged.

For signal OP7:

- If a value of less than 0x005 is received, this will stop any pre-charge that is currently underway.
- If a value greater than or equal to 0x005 and less than 0xFFF is received, this will initiate a pre-charge. The controller will attempt to pre-charge as close as possible to battery voltage.
- If a value of 0xFFF is received, the state of the pre-charge circuit is left unchanged.

Note the use of 0xFFF will leave a device state unchanged. This follows the J1939 convention of “data unavailable”.

3.2 Battery Switch Control Conditions

Although signal OP8 is used to control the battery switch, there are some instances where the controller will not act immediately on these instructions, typically in order to prevent damage to the circuit due to inrush currents.

The battery switch is a solid state device, actually comprised of a number of MOSFETs on the PCB. The battery switch is many times more sensitive to damage through inrush currents than a contactor. If there is a large inrush current, it is possible that this could fuse the MOSFETs, blowing them open circuit and rendering the controller permanently inoperable. In order to avoid this, the low-level driver software for the battery switch includes some conditions that will inhibit opening and closing of the battery switch under certain conditions.

In order to close the battery switch, both of the following must true:

- The difference between Vcap and Vbat must be small in order to minimise inrush current, where small is defined as either Vcap being at least 90% of Vbat, or the difference between Vcap and Vbat being less than 5V
- Vcap must be at least 20V. This condition protects agains reverse battery protection. If the battery is connected in reverse, pre-charge will be impossible and this condition will never be satisfied.

In order to open the battery switch, both of the following must be true:

- No regen current flowing back to the battery
- Drive current being drawn from the battery must be below a safe threshold – currently 50A for Dragon8

3.3 Quick Control DVT Commands for Dragon8 Test Purposes

The following DVT commands can be used to have the DVT produce simulated I-Protocol control commands, in order to control the battery switch and pre-charge circuits. These are useful for test purposes, where the user wants to drive the controller as a pure slave, i.e. without any interaction from BattApp.

The commands work by sending a CAN message on PGN 0x14100 (82176), which is the default PGN for message UO2. This is the message that contains (among others) the signals for OP7 and OP8. This PGN is of type PDU1, which means it can address a single node at a time. This required

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that the default PGN configuration is present in object 0x3F01. The default values can be found in the object dictionary spreadsheet.

The examples given below are for address 0x71, which is the established default address for a standalone controller. However, this should be changed if necessary if the controller address has changed, i.e. the “71” in the message ID should be replaced with the controller’s real J1939 address. The controller address can be confirmed by reading object 0x3E00,1.

The example commands will set control values for outputs that are not to be changed to 0xFFF.

Example command to run the pre-charge and close the battery switch when the pre-charge is complete:

```
can send "0x194171FA 0xFF 0xFF 0xFF 0xFD 0xDF 0xFF 0x00 0x00"
```

Example command to run the pre-charge only:

```
can send "0x194171FA 0xFF 0xFF 0xFF 0xFD 0xFF 0xFF 0x00 0x00"
```

Example command to close the battery switch only:

```
can send "0x194171FA 0xFF 0xFF 0xFF 0xFF 0xDF 0xFF 0x00 0x00"
```

Example command to stop pre-charge and open the battery switch:

```
can send "0x194171FA 0xFF 0xFF 0xFF 0x00 0x00 0x00 0x00 0x00"
```

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4 Mapping Devices to BattApp

In a system that uses Sevcon's vehicle application subsystems, i.e. one that generates drive signals using TracApp rather than using an external VCU, the control signals for the battery switch and pre-charge circuit are generated by BattApp. These signals can be mapped to any digital or analogue output on any device. As described in section 3, the signals can be mapped to either actual contactor outputs, or the virtual outputs that are used in Dragon8 controllers.

The mapping of signals is performed using the standard mechanism as described in the app note "Mapping J1939 items to VehApp". As discussed in this document, the handle for the line contactor is 0x0500, and the handle for the pre-charge control signal is 0x050B. It is these values that should be used in the relevant mappings.

It should be noted that an output signal can be mapped to several physical outputs. For instance, if controlling a dual traction Dragon8 system, the handles for the line contactor and pre-charge outputs can be present in the analogue output signals for both devices.

4.1 Quick Configuration for Standalone Dragon8

For a standalone Dragon8 controller, it is very common to have the line contactor signal mapped to analogue output 8, and the pre-charge signal mapped to analogue 7. Thus, object 0x3E20 would look like this:

```
0x3E20, 1 – 0x71
0x3E20, 2 – 0x????
0x3E20, 3 – 0x????
0x3E20, 4 – 0x0000
0x3E20, 5 – 0x0000
0x3E20, 6 – 0x0000
0x3E20, 7 – 0x0000
0x3E20, 8 – 0x050B
0x3E20, 9 – 0x0500
```

In this example, it is subs 8 and 9 that are important here, as it is here that the handles for the line contactor and pre-charge should be set. This assigns these functions to analogue outputs 7 and 8 on device with address 0x71. Of course, this assumed the local device's address is set to 0x71, which is becoming the common default value for a standalone J1939 device. However, this can be confirmed by checking the address at object 0x3E10,1. Basically, the value at 0x3E20,1 should be set to be the same as that in 0x3E10,1.

Subs 2 and 3 have been left set to unknown values in this example. These can be configured depending on the application, mapping various functions to contactor drives 1 and 2 on the Dragon8. In fact, it is even possible to map the line contactor and pre-charge signals to these outputs as well, if you wish to have an external indication of the state of these devices. If nothing is connected to these outputs, subs 2 and 3 can both be left set to 0x0000.

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5 How BattApp Controls the Line Contactor

For systems that are using the vehicle application subsystems, demands for the line contactor and pre-charge circuits will be generated by BattApp. This section of the document describes how BattApp determined whether the line contactor or battery switch should be open or closed.

5.1 Configuration Quick Reference

The Line Contactor Dropout feature is configured at object 0x2820:

- Sub index 1 should be set to 0x01 if you want to use the line contactor dropout feature, otherwise it should be set to 0x00. If set to 0x00, the remaining sub-indices in this object will be ignored.
- Sub index 2 sets the line contactor dropout time. If there is no drive activity after this timeout, BattApp will request the line contactor to open.
- Sub-indices 3 and 4 are present in the object dictionary, but don't exist in Genus software. Their functions have been replaced by more intelligent line contactor control behaviour, or other configuration objects.

The main control of the line contactor is done at object 0x2821:

- Sub index 1 sets the line contactor open speed. BattApp will not request line contactors to open if the motor on any controller that is protected by a line contactor is above this speed. This is useful for PMAC motors where the back EMC would be excessively large and could damage the controller if the regen path to the battery is interrupted.
- Sub index 2 sets the line contactor close speed. BattApp will not attempt to close the line contactor unless the motor speed is below this level. Again, this is useful for motors with large back EMFs, where the generated voltage at high speed is preventing the line contactor welded checks from taking place. This option stops the line from closing before we can pass the start-up tests.
- Sub index 3 is a bitmap, where we can set bits for motors that are not covered by a line contactor. This inhibits checks such as "line contactor welded" and "line contactor open circuit" for these controllers. If the bit is clear, this means the controller is protected by a line contactor or battery switch. If the bit is set, it means the controller is not protected by a line contactor or battery switch.
- Sub index 4 specifies the maximum arcing voltage for the line contactor. BattApp will not request the line contactor to close until the pre-charge circuit has brought the difference between Vcap and Vbat to below this level.
- Sub index 5 sets the minimum voltage that must be present on Vcap before the line contactor can close. BattApp will not request the line contactor to close if Vcap is below this level.

Sub-indices 1 and 2 should not be left set at zero, as this means the vehicle must be completely stationary in order to either open or close the line contactor. Typically, these speeds would be set to levels that are just below the motor's base speed, i.e. somewhere under the point at which the motor needs to use field weakening to go faster. A speed of 1000rpm for both opening and closing is a suitable default for most applications.

Sub index 4 should also not be left set to zero. It is impossible for the pre-charge circuit to charge the capacitors fully, due to the presence of protection circuits. A small voltage, usually 5V, is a good default value for this object. For applications where vehicle availability and up-time is preferred over protecting the line contactor, then this level can be increased. Should the pre-charge circuit be unable to complete a pre-charge, having a larger maximum arcing voltage would allow the line ocntactor to close and the vehicle to drive.

5.2 Expected Operation

5.2.1 Line Open and Close Conditions

At the heart of the control of the line contactor is a set of conditions that are used to determine if the line contactor should be opened or closed. Essentially, everything else hangs off this.

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The line contactor will be requested to open if the speeds of all motors connected to controllers that are protected by a line contactor is below the level set at object 0x2821,1, and ***any*** of the following conditions are met:

- The line contactor dropout feature is requesting the line contactor to open
- A level 3 fault (severe) or above is set
- The system is not yet initialised

The line contactor will be requested to close if ***all*** of the following conditions are met:

- There is no level 3 (severe) fault or above currently active
- The line contactor dropout feature is not requesting the line contactor to open
- The system has initialised
- The system is in the operational CANopen state
- The speeds of all motors that are connected to controllers that are protected by a line contactor are below the level that is set at object 0x2821,2.

5.2.2 Line Contactor and Pre-charge and Discharge Interaction

Having checked the above conditions and decided whether the line contactor should be open or closed, the system must then invoke a pre-charge or discharge operation, in order to assist with reducing arcing voltages in the contactor, and also to help detect welded line contactors.

If the system has decided that the line contactor needs to be closed, but it is currently open, then it will request a pre-charge. This will bring the capacitor voltage up to a level where the line contactor can be closed without excessive arcing.

If the system has decided that the line contactor needs to be open, but the capacitors are still charged to a level that makes it appear that the line contactor is closed, it will request a discharge. However, there is no discharge output. BattApp actually achieves discharge by requesting motor slaves to enable, but with zero torque demand. The switching action alone at zero torque is sufficient to bring the capacitor down by some measurable amount within around a second, and is used to confirm that the lien contactor is not welded.

5.2.3 Line Contactor State Machine

Figure 2 shows the line contactor state machine as taken from the BattApp design notes:

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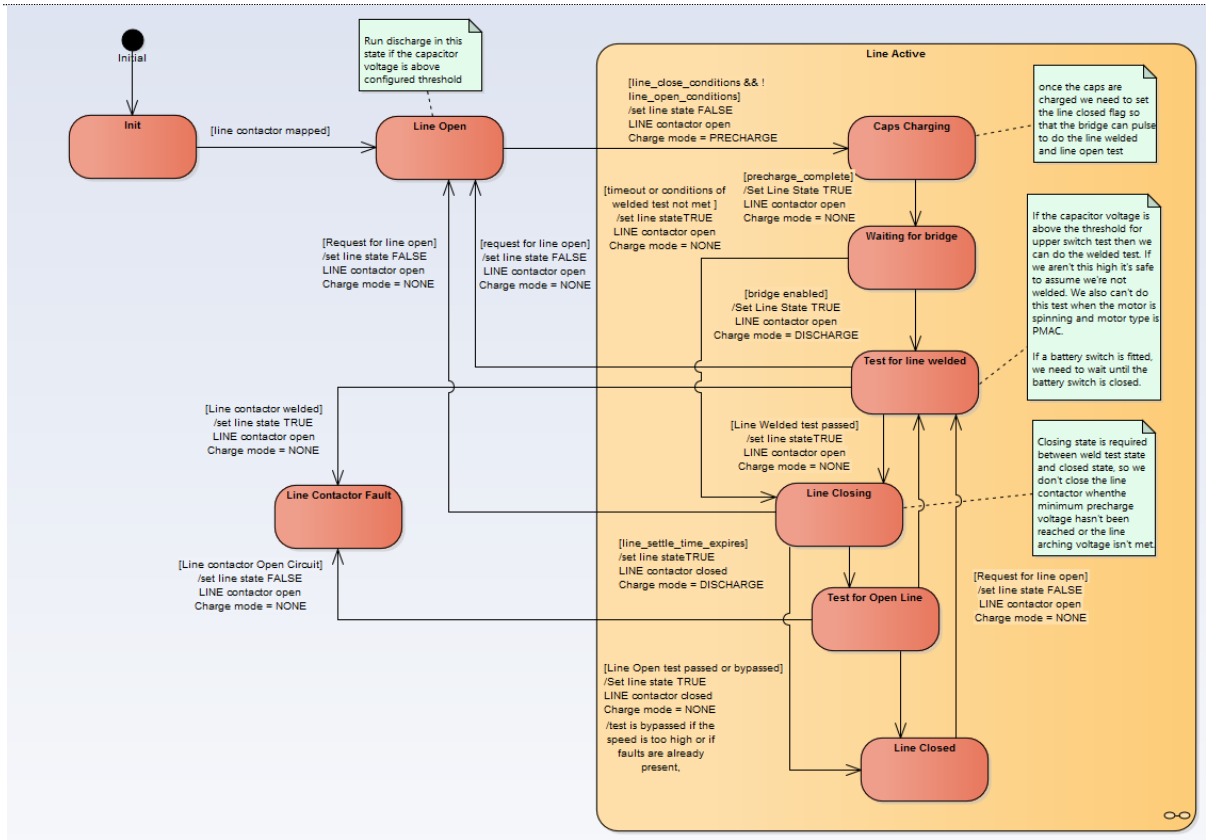


Figure 2 - Line contactor state machine as implemented in Genus software

In the state machine, the guards "line_close_conditions" and "line_open_conditions" are generated from the set of conditions as described in section 5.2.1. These are mutually exclusive, so we shouldn't have both sets of conditions being met simultaneously. However, in case we do, the guards in the state machine are designed so that the open conditions take precedent.

The `Line Open` state is the state that we switch to whenever the line open conditions are met. This transition will result in us requesting the line contactor output is switched off.

If the close conditions are met, we begin to transition through the `Line Active` super state. This involves us moving through the `Caps Charging`, `Waiting for Bridge`, `Test for Line Welded`, `Line Closing`, `Test for Open Line` and `Line Closed` states in turn. The purpose of each state is:

- `Caps Charging` is where we will be pre-charging the capacitor bank. The pre-charge output will be on in this state, however the line contactor will be off.
- When we are satisfied the pre-charge is complete (usually when the difference between Vbat and Vcap falls below the maximum allowed arcing voltage), we make the transition to `Waiting for Bridge`. This is required to allow the Line tests to be conducted. If the bridge is not enabled within 5 seconds it will timeout and skip the line tests. However, we don't allow drive just yet, but we also now have the pre-charge output switched off.
- In the `Test for Line Welded` state the pulsing of the bridge is used to discharge the capacitors to the starting voltage less a threshold set in 0x2821 sub 6, line contactor test disharge threshold. If the capacitors haven't reached this level within the timeout set in 0x2821 sub 7, line contactor test disharge timeout, then the line contactor welded fault is set.
- Once the test has finished we move to the `Line Closing` state. In this state, both the pre-charge and line contactor outputs will be turned on.
- We remain in the `Line Closing` state for 20ms. Although the line contactor output is turned on, it takes a small amount of time before the contactor physically closes. By waiting for 20ms, we avoid allowing the system to drive and prevent us drawing power through a partially closed contactor or, worse, discharging our freshly charged capacitor bank.

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- When the 20ms settle period expired, we move into the `Test for Open Line` state. This uses the same thresholds as the welded test, but the test will check that after the timeout period the capacitor voltage is above the starting voltage less the threshold.
- we move on to `Line Closed`, which is where we will remain, allowing drive to take place. In this state, the line contactor output will be switched on.

5.2.4 Fault Detection

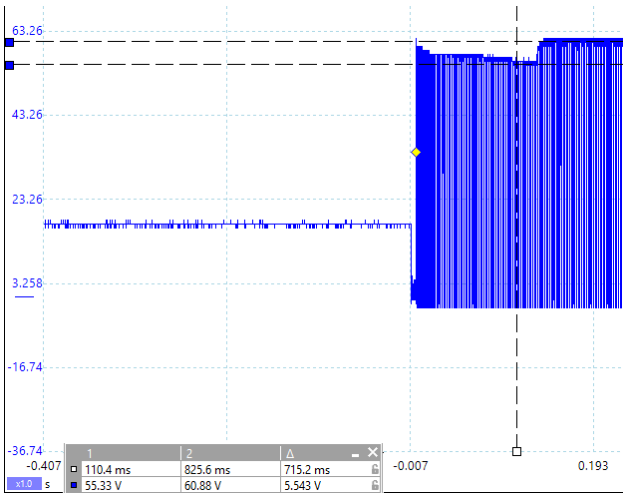
Line contactor fault detection takes place in the `Test for Open Line` and `Test for Line Welded` states.

In `Test for Line Welded` we are looking for “line contactor welded”. In order to do this, we use the pulsing of the bridge to discharge the capacitors to a set level. This level is the voltage when we enter this state less a threshold set in 0x2821 sub 6, line contactor test disharge threshold. If the capacitors haven’t reached this level within the timeout set in 0x2821 sub 7, line contactor test disharge timeout, then the “line contactor welded fault” is set.

In `Test for Open Line`, we are looking for “line contactor open circuit”. This is detected by again using the pulsing of the bridge to try and discharge the capacitors down to a set level. Again, this level is the voltage when we enter this state less a threshold set in 0x2821 sub 6, line contactor test disharge threshold. In this test we do not want the capacitors to discharge. When the timeout set in 0x2821 sub 7, line contactor test disharge timeout, if the voltage is above this threshold the test is passed. If at any time during the test the voltage goes below the threshold, then the “Line Contactor o/c” fault will be set.

Note that differences between Vbat and Vcap might also be triggered by blown power fuses. An incorrect configuration can also cause this, where the system is driving an output that it thinks is the line contactor, but actually isn’t, and in reality the line contactor is still open.

A typical test sequence would look like this:



Once precharging is complete, the bridge starts to pulse (blue channel). The voltage on the capacitors begins to drop until it drops to the threshold shown by the cursor. Then the line contactor closes and the pulsing of the bridge does not discharge the capacitors.

Figure 3 – typical line contactor test sequence

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5.3 BattApp Troubleshooting

Object 0x2700 can be used to help with fault diagnosis in the event that the behaviour of the line contactor is not as expected. Sub-index 1 of this object indicates the state of the line contactor state machine, and sub-index 2 indicates whether we are trying to do a discharge or pre-charge.

If the line contactor is not closing, then it is useful to read object 0x2700,1 to determine the problem:

- If the value of this object is 0x00, then BattApp has decided it shouldn't be closing the line contactor. This is probably because one of the line contactor open conditions, as described in section 5.2.1, has been met. In particular, check to see if there are any faults active, and that the motor speed is below the level that has been set in object 0x2821,2.
- If the value of this object is 0x01, then BattApp has requested a pre-charge, but has decided the capacitors are not sufficiently charged to allow the line contactor to close. Check the measurements for Vbat and Vcap, and ensure the voltage thresholds that are set in objects 0x2821,4 and and 0x2821,5 are set appropriately.
- It is unusual to see this object have a value of 0x02 as it will only stay in this state for 20ms.
- If the value of this object is 0x03, then BattApp will have requested the line contactor to close, and will have set a signal in VehApp's IO manager. It should be possible to see the requested output state being transmitted on I-Protocol. If you can't see it, check your IO mappings at object 0x3E20 and 0x3E21. If you can see it, make sure you're driving the correct output, and that the output, and even the line contactor coil itself, are not broken or damaged.

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